NLP4EMAI

Natural Language Processing for EMAI

Session 1. Introduction

Session outline

NLP in the real world

Language and meaning

Sentiment analysis

The classic NLP pipeline

Evolution of techniques

Activity 1: Tweets analysis

Session objectives

- Understand what **Natural Language Processing (NLP)** is and what problems it tries to solve within AI
- Understand what “sentiment” means when analyzing text from a **computational perspective**
- Distinguish **sentiment analysis** from **emotion detection** within the Emotional AI framework
- Introduce the idea that NLP systems transform human language into **mathematical representations** that enable predictions

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NLP in the real world

Where is NLP today?

Chatbots and assistants

They interpret questions in natural language, enabling intuitive user interactions.

Opinion synthesis

Transforms thousands of reviews into one actionable summary, aiding decision-making.

Social media analysis

Detects trends and public reactions, providing insights into sentiment and discourse.

Discourse analysis

Analyzes digital media and professional settings for understanding communication patterns.

NLP converts human language into machine-processable information, unlocking vast data potential.

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Language and meaning

Why human language is special

- **Semantics:** Words carry meaning that depends on context
- **Pragmatics:** What is said is not always what is meant
- **Implicit Emotion:** Sentiment can be between the lines, with no explicit words
- **Variability:** Synonyms, irony, slang, emojis, that is, the same idea, different expression

Key aspects

Language and meaning

Why human language is special

NLP challenge

They interpret questions in natural language, enabling intuitive user interactions.

Language and meaning

Why human language is special

Key aspect

- **Semantics:** Words carry meaning that depends on context

Consider this example

> **"That concert was sick!"**

The Nuance: In a standard dictionary, "sick" relates to illness (negative). In modern slang/context, it means "amazing" (positive excitement).

NLP Challenge

The model must distinguish between a medical report and a rave review based on the **surrounding words.**

Language and meaning

Why human language is special

Key aspect

- **Pragmatics:** What is said is not always what is meant

Consider this example

> **"Oh, great. Another meeting"**

The Nuance: On the surface, the word "great" is a positive sentiment. However, in this context, it is almost certainly sarcasm expressing frustration or boredom.

NLP Challenge

The machine needs to understand that the **speaker's intent** is the exact opposite of the literal word meaning.

Language and meaning

Why human language is special

Key aspect

- **Implicit Emotion:** Sentiment can be between the lines, with no explicit words

Consider this example

> **"I've been staring at the phone for three hours."**

The Nuance: There are no "emotion words" like sad, lonely, or anxious here. Yet, a human immediately senses anxiety or disappointment.

NLP Challenge

The sentiment is "between the lines." The model must **infer the emotional state** from the situation described, not just from a list of adjectives.

Language and meaning

Why human language is special

Key aspect

- **Implicit Emotion:** Sentiment can be between the lines, with no explicit words

Consider this example

> **"I'm so happy!" vs. "I'm over the moon! 🌙" vs. "This is 'totally' fun... 🙄"**

The Nuance: 1) Direct literal joy. 2) Idiomatic joy + Emoji. 3) Irony (indicated by quotes and the "rolling eyes" emoji).

NLP Challenge

All three use different "tokens" (idioms, punctuation, emojis) to represent or mimic the same semantic field, requiring the model **to be flexible.**

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Sentiment analysis

What does “sentiment” mean in text?

Sentiment analysis identifies the evaluative stance expressed toward an object, service, experience or idea.

⚠ **Sentiment ≠ Emotion**

Sentiment/Polarity

- Positive/Negative/Neutral
- Evaluates an object or experience
- **Example:** "Good film"

Emotion

- Joy, fear, surprise, anger…
- Internal state of the speaker
- **Example:** "I'm so excited"

Sentiment analysis

Types of sentiment analysis

Binary

> *"The film was great." →* ✅ *Positive*

Classifies as positive or negative. The simplest and most common approach in practice.

Sentiment analysis

Types of sentiment analysis

Multiclass (graded)

"It was OK, I guess." → 🟡 *Slightly positive*

Adds categories such as very positive, neutral, very negative. Captures intensity and nuance.

Sentiment analysis

Types of sentiment analysis

Aspect-Based Sentiment Analysis

"Excellent food, terrible service." → 🍽️ + /// 👷 –

One text can contain multiple sentiments about different aspects. Requires extracting WHAT is being talked about and WHAT sentiment is associated.

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From text to prediction, step by step

Sessions 1-3

Raw text

User input

Pre-processing

Cleaning and normalisation

Numeric representation

BoW, TF-IDF, embeddings...

Sessions 4-6

Prediction

+ / − / neutral

Analysis model

ML algorithm

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From text to prediction, step by step

Classic NLP

- Bag of Words
- TF-IDF

Static embeddings

- Word2Vec
- GloVe

Full context

- Transformers
- Attention

Pre-trained models

- BERT
- LLMs

*The **shared goal** of this entire evolution: to model context and meaning ever more accurately.*

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Activity 1: Tweets analysis

Activity 1

Manual analysis of real tweets

No coding — human judgement only

1. **Read each tweet carefully.**
2. **Assign a label:** positive, negative, neutral, or ambiguous.
3. **Discuss** with the person next to you if you disagree.
4. **What would a machine that only counts words do?**
5. **Where would automatic classification be difficult or impossible?**

20
Tweets

Pair work
Format

12 min + discussion
Duration

Think about... 1) Which tweets were hardest to label? 2) Did you and your partner disagree? 3) What clues did you use to decide? 4) What would confuse a word-count model?

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